SAP Note

3360905 - SAP Data Intelligence / SAP Datasphere - ABAP Integration - Performance when using RMS / Replication Flows

Component: CA-DI-IS-ABA (Cross-Application Components > Data Intelligence > Integration Services > Data Intelligence - ABAP Integration), Version: 3, Released On: 24.11.2025

Symptom

You are using / want to use

- the Replication Management Service (RMS) in SAP Data Intelligence
- the Replication Flow feature in SAP Datasphere

You want to understand technical details of the integration of ABAP-based sources in these scenarios and what is influencing the performance of respective data transfer.

Other Terms

SAP Data Intelligence, SAP Datasphere, ABAP Integration, Replication Management Service (RMS), Replication Flow

Solution

1. Introduction

Replication Flows in SAP Datasphere are technically based on the Replication Management Service known from SAP Data Intelligence - in the following it will both be referred to as RMS because the technical integration with the ABAP-based systems is identical.

For the ABAP-based systems, only the scenarios of using SAP LT Replication Server (SLT) to transfer data from tables and the transfer of CDS View data is considered. The integration with ODP data sources works differently and will not be covered.

Important: The information provided in this SAP note assumes a fully patched and up-to-date system. Make sure to regularly check for SAP notes e.g. via our note analyzer report ("3016862 - DMIS Note Analyzers with separated scenarios for ABAP-based Migration and Replication Technology (DMIS2011/DMIS2018/DMIS2020/SAP S/4HANA"). It is especially important that SAP Note "3366842-SAP Data Intelligence - ABAP Integration (DMIS 2018 / S4HANA) - Changing default behaviour of RMS_MAX_PARTITIONS for initial load" is implemented in the ABAP-based source system.

2. How does the ABAP Integration work?

SLT and also the change-data-capture (CDC) engine to provide CDS View data exist as independent products / frameworks. As there are special requirements when integrating with RMS (e.g. to ensure a consistent and resilient data transfer), a buffer framework was introduced in the ABAP-based sources. It is called Resilient Data Buffer (RDB). Within the buffer framework, buffer tables are generated and APIs to read and write the tables are provided.

The basic idea is that the sources (SLT / CDC-engine) push the data to the buffer tables and RMS collects the data from the buffer tables. Within the buffer tables data is grouped into packages and these packages are read by RMS. Once the data is completely processed on RMS side, the respective package is committed and deleted from the buffer table.

3. Factors influencing the performance of the data transfer

The performance is influenced by the following three factors:

- Transfer jobs writing data into the buffer
- RMS connections retrieving data from the buffer
- The size of the buffer table and its size limit

In order to ensure that the buffer tables do not consume to much storage space, the buffer size is limited. Once the buffer size is reached (e.g. as RMS is not processing data fast enough) data will no longer be pushed into the buffer table until some space is freed up again. To achieve the best performance possible, it is therefore necessary that the jobs (from SLT or the CDC engine) push data with the same speed into the buffer table as RMS is able to read, process and commit the data packages.

4. Optimizing performance

If you want to optimize performance, it is important to monitor the state of the buffer tables.
If your ABAP-based source system is using SAP S/4HANA or SAP S/4HANA Foundation, the transaction to monitor buffer tables is called DHRDBMON. In case of a SLT system with the DMIS 2018 addon, the transaction is called LTRDBMON.
Depending on the status of data packages in the buffer, the following actions can be taken:

1. The buffer table is empty / only a few packages are visible from time to time and are immediately set to "In Process". In case there is still data to process from source side (e.g. as it is still the initial load), this means RMS is retrieving the data

from the system faster than the source is pushing data into the buffer.
    - In case of SLT: Increase the number of transfer jobs in the respective configuration and for the respective table in transaction LTRC.
    - In case of CDS Views via the CDC engine: Increase the number of transfer jobs, see SAP Note "3223735 - SAP Data Intelligence - ABAP Integration - Job Settings (Transaction DHCDCSTG / Table DHCDC_JOBSTG)".
    - **Note 1:** The upper limit of transfer jobs working on one object at the same time during the initial load phase is the number of access plans. To increase the number of access plans, the calculation phase needs to be executed again with different settings.
        - SLT: More details can be found in the SLT performance guide on help.sap.com/slt.
        - CDS: It is currently not possible to influence the number of access plans created by the system.
    - **Note 2:** The upper limit of transfer jobs working on one object at the same time during the replication phase depends on the source.
        - SLT: Settings for parallel processing need to be done in transaction LTRS before sending the request for data transfer from RMS side (see SLT performance guide on help.sap.com/slt). The upper limit of transfer jobs is the number of access plans created based on the LTRS settings.
        - CDS: For delta processing the CDC engine providing CDS data is not using the concept of access plans. Hence, parallel processing only depends on the number of transfer jobs (see SAP Note "3223735 - SAP Data Intelligence - ABAP Integration - Job Settings (Transaction DHCDCSTG / Table DHCDC_JOBSTG)").

2. The buffer table is full. The status of the packages is "Ready". This means the source is pushing data faster to the buffer than RMS is retrieving the data from it.
    - Try increasing the number of connections in the respective Replication Flow. The maximum amount of connections used at the same time for one object is the number of partitions.
    - **Note:** For delta replication, the number of partitions (and consequently the number of connections used to transfer data of an object) is currently fixed to one.

3. The buffer table is full. The status of the packages is "In Process". The source is pushing data and RMS is immediately picking it up. Most likely a higher throughput could be achieved if the buffer size is increased as source and RMS would allow more parallel processing.
    - In transaction DHRDBMON / LTRDBMON choose the "Expert Function" button. Enter the buffer table name via the F4 help and choose the "Change Settings" radio button. You can freely modify the maximum amount of records to be stored in the buffer table and the individual package size. The changed settings will be active immediately. It is recommended to **not** change the individual package size and to choose a multiple of the package size for the buffer table size. For example the package size is 51200 records and the current buffer size is 153600 (three packages). So the new buffer size could be 307200 (six packages).

5. Package Size Limit on ABAP Side and RMS Handling

For the ABAP Integration scenario (SLT and CDS/CDC), the size of an individual data package sent to RMS is calculated on the ABAP side based on the memory footprint of the records in the buffer table. The package size calculation uses the row length of the buffer table (TABLEN) and the configured number of records per package.

It is recommended that the resulting package size does not exceed approximately 30 MB. This is the limit used by RMS to ensure stable processing and avoid connection resets, timeouts, or memory pressure situations.

The 30 MB limit refers to the serialized payload prepared by ABAP. The wire protocols used for transmission may add a small amount of protocol overhead. This is acceptable as long as the resulting size is not significantly above ~30 MB, for example:

Package size: 164,835 rows
Row length (TABLEN): 202 bytes
Calculated payload: 164,835 * 202 bytes = 33,296,670 bytes ≈ 31.75 MB

This exceeds the recommended threshold and may lead to unstable behavior during transfer.

If the calculated package size approaches or exceeds ~30 MB, reduce the package size in transaction DHRDBMON / LTRDBMON. In DHRDBMON / LTRDBMON → Expert Functions → Change Settings (select the buffer table via F4), you can adjust the individual package size; changes take effect immediately.

6. Additional Information

RMS shows a number of partitions on its monitoring UI. Partitions are used in RMS to determine the amount of parallel connections for an object. The number of available partitions in case of the initial load from an ABAP-based source is sent from the source to RMS. For the ABAP integration it is equal to the number of access plans available in the source. If the number of partitions should be limited, parameter RMS_MAX_PARTITIONS (see SAP Note "3260185 - SAP Data Intelligence / SAP Datasphere - ABAP Integration - Runtime Parameters (DHBAS_RUNTIME / LTBAS_RUNTIME)") can be used.

The default buffer table size is three packages per partition (per available access plan). Nevertheless, by default, the buffer table size is limited to thirty packages (so as if there would be ten partitions / access plans).